1. Project Overview

The project focuses on designing and developing a motorized optical fibre cable drum system. The system will be capable of controlled cable deployment and automatic cable retrieval using a BLDC motor and a direct-coupling mechanism.

The main objective is to determine the required motor torque, power, and controller specifications through practical measurements rather than relying only on theoretical assumptions.

2. Current Design Specifications

Parameter	Specification
Optical fibre cable weight	32 kg
Drum weight	3 kg
Total rotating mass	35 kg
Drum core diameter	200 mm
Drum width	300 mm
Flange diameter	300 mm
Drum shaft diameter	20 mm (provisional)

Required drum speed	50 RPM
Motor type	BLDC
Battery	22.2 V LiPo (6S)
Transmission	Direct coupling
Gearbox	Not required in current design
Ground surface	Soil

3. Research Objective

The research will focus on the following engineering parameters:

1. Determination of actual cable pulling force.
2. Calculation of required drum torque.
3. Calculation of motor mechanical power at 50 RPM.
4. Selection of a suitable BLDC motor.
5. Selection of a compatible bidirectional motor controller.
6. Design of the shaft, bearings, and flexible coupling.
7. Development of a controlled freewheel mechanism.
8. Structural analysis using Fusion 360 and ANSYS.

4. Two Calculation Approaches

Option 1: Practical Experimental Calculation

KERN digital force gauge SAUTER FL 20, 427,50€

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- Use a spring balance or digital force gauge.
- Place the optical fibre cable on the actual soil surface.
- Measure the initial pulling force (breakaway force).
- Measure the continuous pulling force.
- Record the maximum pulling force under controlled test conditions.
- Use the measured force to calculate the required motor torque.

Option 2: Software-Based Calculation

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- Develop a 3D CAD model in Fusion 360.
- Apply the estimated mechanical loads.
- Analyse shaft stress and deformation.
- Evaluate bearing loads.
- Verify the design using ANSYS simulation.

5. Preliminary Engineering Calculations

Drum core radius:

$$r=\frac{200}{2}=100\text{ mm}=0.1\text{ m}$$

Maximum flange radius:

$$r_{\max}=\frac{300}{2}=0.15\text{ m}$$

Required torque:

$$T=F_{\text{pull}}\times r_{\text{effective}}$$

Mechanical power:

$$P=\frac{2\pi NT}{60}$$

For an illustrative pulling force of 100 N at a 150 mm effective radius:

$$T=100\times0.15=15\text{ Nm}$$

With a preliminary design factor of 2:

$$T_{\text{design}}=30\text{ Nm}$$

At 50 RPM:

$$P=\frac{2\pi\times50\times30}{60}$$

$$\boxed{P=157.1\text{ W}}$$

These values are preliminary examples, not experimentally validated motor requirements.

6. Proposed Research Methodology

1. Practical force measurement
   Measure cable pulling force on soil.
2. Torque and power calculation
   Calculate required torque at different drum radii.
3. BLDC motor selection
   Select motor based on torque, RPM, voltage, and current.
4. Controller selection
   Select a bidirectional BLDC controller.
5. Mechanical CAD design
   Model drum, shaft, bearings, and coupling.
6. Simulation and validation
   Verify structural strength and system performance.
7. Prototype testing
   Test cable deployment, retrieval, speed control, and braking.

7. Expected Research Outcome

The research is expected to produce:

- An experimentally validated motor torque requirement.
- A suitable BLDC motor and controller specification.
- A mechanically safe drum shaft and bearing arrangement.
- A controlled cable deployment and retrieval mechanism.
- A functional test-bench prototype.

8. Immediate Next Step

The first practical activity will be measuring the actual optical fibre cable pulling force using a calibrated digital force gauge or spring balance.

The cable manufacturer's maximum allowable pulling tension and minimum bend radius must be respected during testing. The fibre itself should not be used as an unrestricted pulling member.

Research approach: Practical Experimentation → Mathematical Calculation → CAD Design → FEA Simulation → Prototype Validation.